

02443 Stochastic Simulation

Day 5: Markov Chains

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The queueing system

Consider the queueing model from the previous exercises with a Poisson arrival process and exponentially distributed service times. This is an example of a Markov process.

We simulated the queueing system until it reached (stochastic) steady state and characterized this steady state:

- What is the distribution of occupied servers?
- What is the rejection probability?

Logic

We used stochastic simulation to obtain steady-state statistics.

Later: We shall reverse this logic, i.e., use the steady-state distribution of a stochastic process to generate random variables.

Definition of a Markov chain

A Markov chain is a discrete time stochastic process with a discrete state space, whose evolution only depends on the current state of the process.

Components

- A finite or infinite discrete state space S
- A discrete time index T
- A transition kernel (specifically a Markov kernel)

For example, we could have a stochastic process, $\{X_t\}_{t \in T}$, with $S = \{0, 1, \dots, 6\}$, $T = \mathbb{N}_0$, and a Markov kernel $\mathbf{P} = \{p_{ij}\}_{i,j \in S}$ with elements:

$$p_{ij} = \mathbb{P}(X_{t+1} = j | X_t = i), \quad t \in T, \quad i, j \in S.$$

The element p_{ij} gives the probability of the process transitioning to state j when it is in state i at time t .

Examples of Markov chains

Notice that in this case:

- The transition probability only depends on X_t
- The transition probability is independent of t

The first condition is called the Markov property and applies to all Markov processes. The second condition implies that $\{X_t\}$ is a time-homogeneous process.

Examples of Markov chain models

- (Discretised) cloud cover on successive days in January
- Number of cars in stock at a car dealer at the beginning of a day
- Number of communication packets in a buffer at the beginning of a transmission slot

Distributions in the Markov chain

Consider the Markov process $\{X_n\}_{n \in T}$ with state space S , time index T , and transition probability matrix P .

Marginal distribution

The marginal distribution of the Markov chain at time/step n is the distribution $p(n) = \{p_i(n)\}_{i \in S}$, where

$$p_i(n) = \mathbb{P}(X_n = i).$$

You can show that

$$p(n) = p(n-1)P = p(0)P^n.$$

Stationary distribution

A stationary distribution of the Markov chain is a distribution $\pi = \{\pi_i\}_{i \in S}$ satisfying the equation

$$\pi = \pi P,$$

i.e., the marginal distribution remains unchanged as time progresses.

Limiting distribution

The limiting distribution, when it exists, is the distribution to which the marginal distributions converge. Any limiting distribution is a stationary distribution, but the converse is not true.

The embedded Markov chain

Consider again the queueing system with a Poisson arrival process and exponentially distributed service times.

Assume that the system is not idle nor at maximum capacity (m). At any given time, the next event will either be a customer arriving and entering the system or a customer finishing service and leaving the system.

We are interested in calculating the probability that the next event is an arrival rather than a departure (and vice versa). In other words, if we denote the number of customers in the system at time t by N_t , what are the probabilities that the process $\{N_t\}$ will transition to the states with one more or one fewer customer.

The embedded Markov chain (2)

We can derive that

- ① The time until the next arrival is exponentially distributed
- ② The time until the next departure is exponentially distributed
- ③ The time until the next event is exponentially distributed

using the independence of the two times and the memoryless property: For $A \sim \exp(\lambda)$

$$\mathbb{P}(A > \Delta + t | A > t) = \frac{\mathbb{P}(A > \Delta + t, A > t)}{\mathbb{P}(A > t)} = \frac{\mathbb{P}(A > \Delta + t)}{\mathbb{P}(A > t)} = \frac{e^{-\lambda(\Delta+t)}}{e^{-\lambda t}} = e^{-\lambda \Delta}.$$

If $0 < k < m$ customers are in the system and in service, each having an exponentially distributed service time $S_i \sim \exp(\mu)$, the time until the next departure is

$$D = \min_{i \in \{1, \dots, k\}} \{S_i\}.$$

The embedded Markov chain (3)

The survival function of D is given by

$$\mathbb{P}(D > t) = \mathbb{P}(S_1 > t, S_2 > t, \dots, S_k > t) = \prod_{i=1}^k \mathbb{P}(S_i > t) = \prod_{i=1}^k e^{-\mu t} = e^{-k\mu t},$$

and thus $D \sim \exp(k\mu)$. Finally, the time until the next event is given by $E = \min(A, D)$ with survival function

$$\mathbb{P}(E > t) = \mathbb{P}(D > t, A > t) = \mathbb{P}(D > t)\mathbb{P}(A > t) = e^{-\lambda t} e^{-k\mu t} = e^{-(\lambda + k\mu)t},$$

and thus $E \sim \exp(\lambda + k\mu)$. Furthermore,

$$\mathbb{P}(E = A) = \mathbb{P}(A < D) = \int_0^\infty \mathbb{P}(A < D | A = t) f_A(t) dt = \int_0^\infty e^{-k\mu t} \lambda e^{-\lambda t} dt = \frac{\lambda}{\lambda + k\mu}$$

$$\mathbb{P}(E = D) = 1 - \mathbb{P}(E = A) = \frac{k\mu}{\lambda + k\mu}.$$

Important result

The key observation is:

The transition probabilities do not depend on time. In other words, the transition probabilities in such a Markov chain do not depend on how long it takes the process to transition.

Therefore, we have the following result:

We can simulate the sequence of states without simulating the transition times. We can simulate the transition times afterwards, as long as we know the sequence of states.

Global balance equations

Suppose that a Markov chain has a stationary distribution $\pi = (\pi_1, \pi_2, \dots, \pi_N)$. By definition, a stationary distribution satisfies

$$\pi = \pi P.$$

Looking at a single state j , this equation can be written as

$$\pi_j = \sum_{i \in S} \pi_i p_{ij}.$$

Interpretation

- π_j is the long-run probability of being in state j .
- $\pi_i p_{ij}$ is the long-run probability flow from state i to state j .

The global balance equations:

$$\sum_{i \in S} \pi_i p_{ij} = \sum_{i \in S} \pi_j p_{ji}, \forall j \in S.$$

It implies that for each state j :

flow into state j = flow out of state j .

Local balance equations

A stronger condition is obtained by requiring balance for each pair of states:

$$\pi_i p_{ij} = \pi_j p_{ji}, \quad i, j \in S.$$

This condition is known as the local balance equations or the detailed balance equations.

Interpretation

For every pair of states:

$$\text{flow from } i \text{ to } j = \text{flow from } j \text{ to } i.$$

A Markov chain satisfying the local balance equations is called reversible.

Important

Local balance implies global balance, but the converse is not generally true.

A small numerical example

Consider the Markov chain with transition matrix

$$P = \begin{pmatrix} 1-p & p & 0 & 0 \\ q & 0 & p & 0 \\ 0 & q & 0 & p \\ 0 & 0 & q & 1-q \end{pmatrix}.$$

Suppose the initial distribution is

$$p(0) = \left(\frac{1}{3}, 0, 0, \frac{2}{3}\right).$$

The distribution after one step is

$$p(1) = p(0)P.$$

Performing the matrix multiplication gives

$$p(1) = \left(\frac{1-p}{3}, \frac{p}{3}, \frac{2q}{3}, \frac{2(1-q)}{3}\right).$$

Two-step transition probabilities

Recall that

$$\mathbf{p}(2) = \mathbf{p}(0)\mathbf{P}^2.$$

We therefore first compute

$$\mathbf{P}^2 = \begin{pmatrix} (1-p)^2 + pq & (1-p)p & p^2 & 0 \\ q(1-p) & 2pq & 0 & p(1-q) \\ q^2 & 0 & 2pq & p(1-q) \\ 0 & q^2 & (1-q)q & (1-q)^2 + pq \end{pmatrix}.$$

The matrix entry in row i and column j gives the probability of moving from state i to state j in exactly two steps.

Distribution after two steps

Using

$$\mathbf{p}(2) = \mathbf{p}(0)\mathbf{P}^2,$$

we obtain

$$\mathbf{p}(2) = \left(\frac{1}{3}, 0, 0, \frac{2}{3}\right) \mathbf{P}^2.$$

After multiplication,

$$\mathbf{p}(2) = \left(\frac{(1-p)^2 + pq}{3}, \frac{(1-p)p}{3}, \frac{4pq}{3}, \frac{2p(1-q)}{3}\right).$$

In general,

$$\mathbf{p}(n) = \mathbf{p}(0)\mathbf{P}^n.$$

Thus, once the transition matrix is known, the distribution at any future time can be determined.

Local balance for the example

Applying the local balance equations gives

$$\pi_i p = \pi_{i+1} q \implies \pi_{i+1} = \frac{p}{q} \pi_i.$$

Applying this recursively yields

$$\pi_i = \left(\frac{p}{q}\right)^{i-1} \pi_1, \quad i = 1, 2, 3, 4.$$

Using the normalization condition

$$\sum_{i=1}^4 \pi_i = 1,$$

we obtain

$$\pi_1 = \left(\sum_{i=1}^4 \left(\frac{p}{q}\right)^{i-1} \right)^{-1}.$$

Thus, the entire stationary distribution is determined.